

EXPERIMENT-16

1'S COMPLIMENT.

AIM: To write assembly language to find 1's COMPLIMENT by using 8085 microprocessor simulator.

ALGORITHM:

1. Load value into A.
2. Complement all bits.
3. Store result at 8001H.
4. Halt program.

PROGRAM:

LDA 8000 - Load input value.

CMA - complement all bits

STA 8001 - Store complemented value.

HLT - Stop program.

Input:

Address	data
8000	6

Output:

Address	data
8001	249

Result: This program was executed successfully by using 8085 microprocessor simulator.

Registers			Flag
<i>A</i>	F9		<i>S</i> 0
<i>BC</i>	00	00	<i>Z</i> 0
<i>DE</i>	00	00	<i>AC</i> 0
<i>HL</i>	00	00	<i>p</i> 0
<i>PSW</i>	00	00	<i>C</i> 0
<i>PC</i>	42	08	
<i>SP</i>	FF	FF	
<i>Int-Reg</i>	00		

Load me at

```
1 LDA 8000
2 CMA
3 STA 8001
4 HLT
```

Decimal - Hex Conversion

Decimal	Hex
0	0
→ To Hex	← To Dec

I/O Ports

Memory

8000

-

+

06

 Update Memory

 Data
  Stack
  KeyPad
  **Memory**
 I/O Ports

Start 8001 OK

Address (Hex)	Address	Data
1F41	8001	249
1F42	8002	0
1F43	8003	0
1F44	8004	0
1F45	8005	0
1F46	8006	0
1F47	8007	0
1F48	8008	0
1F49	8009	0
1F4A	8010	0
1F4B	8011	0
1F4C	8012	0
1F4D	8013	0
1F4E	8014	0
1F4F	8015	0
1F50	8016	0

Line No	Assembler Message
0	Program assembled successfully